

VDA Signal Coherence

Laboratory Notebook

A claim-by-claim reconstruction of behavior, visual and memory routing, representation, and causal use

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Purpose of this version. The original 41-page manuscript is preserved unchanged as `VDA_Set_Size.pdf`. This separate notebook reorganizes the same admitted evidence into a short Herman–Morgan-style result sequence, then adds the full experimental logic, negative results, measurement definitions, interruption record, and next-decision criteria. It is a readability and evidential-hierarchy reconstruction, not a silent replacement of the scientific record.

Current verdict

The experiments show that at least one trained VDA checkpoint uses spatially targeted routing causally: suppressing the true change location impairs detection and boosting it rescues detection, while the same operation at wrong locations does not produce the rescue. The current evidence does *not* yet establish the broader theory that attention generally increases signal coherence, nor a monotonic law that finer image discretization makes attention more necessary. The missing centerpiece is a replicated causal chain from routing, through target representation and actor evidence, to sensitivity-corrected behavior.

Contents

How to read this notebook

The document has two reading speeds. Part I contains the decisive scientific sequence in the order a reader needs it: task, behavior, source-separated dynamics, causal intervention, then scaling tests. Part II records each experiment as a notebook entry with its prediction, frozen design, result, admitted claim, withheld claim, and next action. Part III defines the attention measures and preserves the production and provenance record.

The sequence is deliberately modeled on Morgan, Albanna, and Herman’s recurrent-vision-transformer paper [?]: establish behavior before reading internal maps; align maps to meaningful trial events; perturb the same quantity at the correct and incorrect places and times; and ask whether alternative architectures or controls reproduce the effect. The present notebook strengthens that logic by separating visual and recurrent-memory attention, decomposing sensitivity from criterion, treating independent training seeds as the inferential unit, and recording null, mixed, blocked, and interrupted cells explicitly.

EVIDENCE LEVEL	WHAT IT CAN ESTABLISH	WHAT IT CANNOT ESTABLISH
Engineering integrity	A checkpoint or cache is complete, finite, hash matched, and reproducible	Behavior, attention, mechanism, or a theory
Held-out behavior	Psychometric, chronometric, false-alarm, d' , and criterion effects at a frozen checkpoint	Which internal variable caused the effect
Routing description	Where visual or memory attention mass goes at a named frame	Cleaner target representation or policy use
Representation probe	Information is linearly available in a named state	That the trained actor uses it or that attention created it
Causal intervention	Changing a named routing variable changes matched behavior with spatial controls	Population generality, biological equivalence, or the full mediation chain
Independent seeds	Variation over separately trained models	Generality beyond sampled tasks, architectures, and training protocol

Table 1: Evidence levels are cumulative. Training metrics, curriculum values, periodic checkpoints, and GPU/process health remain engineering evidence only.

Evidence boundary. Throughout this notebook, “attention” means the model’s learned routing weights. “Signal coherence” is a functional hypothesis about alignment and propagation of task-relevant information, not oscillatory or spectral neural coherence. No routing tensor is identified with FEF, SC, LPFC, or biological working memory.

Part I — The decisive result sequence

1 The question Herman–Morgan leaves open

Morgan, Albanna, and Herman demonstrated a compact progression: a recurrent vision transformer shows cue-dependent psychometric and response-timing effects; its attention shifts over cue, delay, array, and change; forcing attention toward the changed location improves declarations while forcing it elsewhere hurts; and several alternative training or feedback schemes fail to reproduce the whole pattern [?]. That paper therefore supports a model-level claim about top-down priority allocation. It does not directly measure a quantity called signal coherence.

The VDA hypothesis is more demanding. Attention should not merely become sharper or move to a target. It should improve the functional chain by which target-consistent evidence survives competition and reaches the decision policy:

target-aligned visual/memory routing → **separable target representation** → **stronger actor evidence** → **higher d' / lower threshold / faster correct response**

The present VDA archive directly observes the first and fourth terms, probes some recurrent states, and causally changes routing and behavior at a few checkpoints. It does not yet observe and intervene on every link on the same held-out trials. Consequently, the correct current claim is that the coherence account is plausible and sharply falsifiable, not proven.

HERMAN–MORGAN STEP	VDA COUNTERPART	IMPROVEMENT REQUIRED HERE
Behavior and response time	Psychometrics plus discrete response frame	Add false-alarm arms, d' , criterion, and nonsaturated calibration
Temporal attention maps	Cue/delay/change-aligned routing	Always split visual keys and recurrent-memory keys; define the matrix reduction exactly
Correct versus wrong perturbation	True-change, cued-wrong, and neutral biases	Add visual-only, memory-only, norm-matched, and time-specific interventions
Alternative models	Cross-attention, affine, width, and decay comparisons	Parameter-match architectures and replicate complete interactions over training seeds
Functional interpretation	Signal-coherence hypothesis	Demonstrate routing → representation → actor → behavior mediation

Table 2: The rewrite retains Herman–Morgan’s readable evidentiary escalation while tightening measurement and inference.

2 Entry LN-00: Frozen hypothesis and failure conditions

STATUS	pending	EVIDENCE CLASS	Theory audit
QUESTION	Does attention increase the coherence of target-consistent signals relative to competitors?	PRIMARY UNIT	Independent trained checkpoint; matched held-out trial

OPERATIONAL PREDICTION. On a decision-relevant frame, a valid cue or correctly targeted intervention should increase target-versus-distractor routing, improve the separability and temporal stability of target representations, strengthen change evidence at the actor, and improve d' or threshold without merely liberalizing criterion. The same operation at a cued-but-wrong or neutral location should not reproduce that chain.